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Submitted via: www.regulations.gov

Re: Lennox Comments on DOE Request for Information on Test Procedure and Energy Conservation Standards for Consumer Products; Consumer Air Cleaners (Docket No. EERE-2021-BT-STD-0035 and EERE-2021-BT-TP-0036).

Lennox International Inc. (Lennox) hereby submits comments on the Request for Information concerning the *Energy Conservation Program: Test Procedure and Energy Conservation Standards for Consumer Products; Consumer Air Cleaners* published by the U.S. Department of Energy (DOE) in the Federal Register on January 25, 2022 (87 Fed. Reg. 3,702)(the “RFI”).

Lennox is a leading provider, based in the United States, of climate control solutions for heating, air-conditioning and refrigeration equipment (HVACR). Lennox is a publicly traded company, has thousands of employees, and offers a broad range of HVACR products to the marketplace. This industry is an important source of American jobs and provides equipment vital to the health and wellbeing of consumers and the preservation of food. Related to issues discussed in the RFI, Lennox offers air cleaning devices that are part of a central air conditioning or furnace system. Lennox offers the following general comments on the RFI.

Scope and Air Cleaner Definition: Lennox strongly agrees with DOE’s proposal that air cleaners associated with central air conditioning and furnace systems be excluded from the scope of a DOE consumer air cleaner efficiency standard.

Lennox strongly agrees with the DOE proposal that air cleaners associated with central air conditioning and furnace systems be excluded from the scope of a consumer air cleaner energy efficiency standard. DOE proposed, in the September 2021 notice of proposed determination (NOPD) that preceded this RFI, to define a consumer air cleaner as a consumer product that *excludes* central air conditioners and furnaces. In particular, DOE proposed to define a consumer air cleaner as a product that:

- (1) Is a self-contained, mechanically encased assembly;
- (2) Is powered by single-phase electric current;
- (3) Removes, destroys, or deactivates particulates and microorganisms from the air; and

- (4) Excludes products that destroy or deactivate particulates and microorganisms solely by means of ultraviolet (“UV”) light without a fan for air circulation; and
- (5) Excludes central air conditioners, room air conditioners, portable air conditioners, dehumidifiers, and furnaces as defined in 10 CFR 430.2.

See NOPD at 86 FR 51,632, as cited in the RFI at 87 FR 3,707. Lennox strongly agrees with DOE’s proposal that whole-home air cleaners associated with central air conditioning and furnace systems be excluded from coverage under a consumer air cleaner efficiency standard. These whole-home air cleaners use the fan from a central air conditioner or furnace to move air through the air cleaner, and the associated energy is captured in the efficiency metrics for central air conditioner and furnace systems. Additional regulation to include air cleaners that are part of a central air conditioning or furnace system would be duplicative and increase manufacturer and consumer burden without providing benefit, since the fan energy use is already addressed by other DOE regulations. While some premium consumer air cleaners that are part of a central air conditioning and or furnace system may have additional features, e.g., monitoring and communication capability with a low level of energy consumption, this energy use is de minimis and the majority of systems require no additional power source or energy consumption beyond that which is already captured in the central air conditioner or furnace metrics. Furthermore, as DOE notes, leading air cleaner standards, including those standards of AHAM and ENERGY STAR, focus on “portable” units or those which can be “moved from room to room.” (RFI at 3,707). This focus on portable units – but not central air conditioners and furnaces – is appropriate for this consumer air cleaner rulemaking.

Lennox also agrees with DOE excluding UV lights from the definition of consumer air cleaner. DOE notes that UV lights would fall under “lamp” regulations. (RFI at 3,707). To avoid duplicative regulatory approaches, Lennox supports DOE excluding UV lights from the definition of portable air conditioner and thus excluding them from this rulemaking’s scope of coverage.

Test Procedure: this RFI’s test procedure approach appropriately excludes central air conditioners and furnaces.

As part of this rulemaking, DOE has “consulted existing definitions and classifications of consumer air cleaners developed by AHAM—the industry trade group for consumer air cleaners—and the ENERGY STAR Program.” (RFI at 3,707). AHAM AC-1-2020 may be an appropriate test method for consumer air cleaners that remove particulate matter and meet the DOE proposed consumer air cleaner definition, but this test procedure is not applicable to air cleaners that use the airflow from fans of central air conditioning and furnace systems to operate. Likewise, the ENERGY STAR program focuses on portable units.

The industry standard test procedure ASHRAE 52.2-2017 (“Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size” cited in the RFI at 87 FR 3,709) defines the method of test for evaluating general air-cleaning performance as a function of particle size. This industry standard measures the Minimum Efficiency Reporting Value (MERV) rating of these products. At this time, Lennox finds ASHRAE 52.2-2017 standard methodology acceptable for air cleaners that remove particles. However, as noted

above, central air conditioners and furnaces should be excluded from the definition of air cleaners and this rulemaking's scope of coverage.

While Lennox understands DOE is considering other methodologies that measure the removal efficacy of microorganisms and VOCs, these pollutants present complex issues DOE must consider before proceeding with a test procedure or standard. While other methodologies are available that consider microorganisms and VOCs, a universal standard for testing their removal efficiency is not currently available, which challenges the ability to compare product performance and determine the representativeness of efficiency values. Lennox understands AHAM is currently working to include microorganisms as a new contaminants in its air cleaner standard and DOE should await the outcome of this AHAM process before applying microorganism standards for the scope of products DOE has proposed addressing in this RFI.

DOE must maintain the consumer utility of air cleaners when promulgating new standards and must ensure that any new standards are economically justified.

DOE must proceed with caution to ensure the primary function of consumer air cleaners, to improve indoor air quality and associated health considerations, is not inhibited by test procedures or energy conservation standards that limit technology and innovation. DOE must carefully consider and ensure that any standards regarding consumer air cleaners provide significant energy savings and are economically justified. (See e.g., RFI at 3,704 citing 42 U.S.C. 6295(o)(3)). As the RFI notes, DOE weighs the consumer "utility" for the product when considering standards (see e.g., RFI at 3,705 and 3,706), and DOE must carefully consider such consumer utility for any new consumer air cleaner standards.

Cumulative regulatory burdens on HVAC manufacturers provide more grounds for excluding central air conditioners from the scope of coverage.

The HVAC industry is facing multiple evolving standards that impact our business. These include increasingly stringent efficiency standards that are scheduled to take effect over the next several years, transitioning to lower emission refrigerants and development of new products that support national decarbonization efforts. Such standards include:

- 2023 DOE Energy Conservation Standard (ECS) change for Central Air Conditioners,
- 2023 DOE ECS change for Commercial Air Conditioners
- 2023 DOE ECS for Commercial Warm Air Furnaces
- EPA Phase-down to lower GWP refrigerants to meet the AIM Act objectives.
- DOE ECS Furnace Standards rulemaking
- National and Regional Cold Climate Heat Pump Specifications
- DOE ECS for Three Phase below 65,000 Btu/h
- DOE Test Procedure for VRF Systems
- EPA Energy Star 6.0+ for Residential HVAC
- EPA Energy Star 4.0 for Light Commercial HVAC
- DOE Test Procedure for Commercial Warm Air Furnaces
- DOE Test Procedure for Walk-In Coolers and Freezers

- DOE ECS for Walk-In Coolers and Freezers

These changes within the HVACR will significantly increase the energy efficiency of these products and reduce emissions. These changes in themselves will drive more efficient design without unnecessary further regulation of accessory components such as air cleaners. The cumulative regulatory burden of these evolving standards provides additional support for DOE's proposal to exclude central air conditioners and furnaces from the scope of coverage of consumer air cleaner regulations.

In summary, Lennox strongly agrees with DOE's exclusion of air cleaners that are part of a central air conditioning or furnace system from the definition of a consumer air cleaner. DOE must carefully consider and ensure any standards regarding consumer air cleaners provide significant energy savings and are economically justified. Further, DOE should proceed with caution to ensure the primary function of consumer air cleaners, to improve indoor air quality and associated health considerations, is not inhibited by test procedures or energy conservation standards that limit technology and innovation.

Sincerely,

A handwritten signature in cursive script, reading "David Winningham".

Dave Winningham,
Sr. Engineering Manager, Regulatory Affairs